

CS228 : Logic for Computer Science

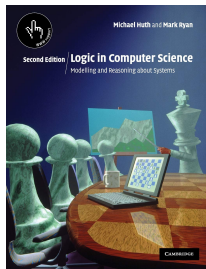
Lecture 1: First Module on Logic and Reasoning

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Logistics

- ▶ Course webpage: <https://www.cse.iitb.ac.in/~akshayss/courses/cs228>
- ▶ Piazza invites sent!
- ▶ Textbook: For the first part of the course, we will use Huth and Ryan's book on **Logic in Computer Science**.



Module 1: Logic, the language/calculus for reasoning

A whole zoo of symbolic logics out there!

We will go from simple to more complicated

- ▶ Propositional Logic
- ▶ Predicate Logic
 - ▶ First order logic
 - ▶ Monadic second order logic
 - ▶ Higher order logics
- ▶ Temporal logics
- ▶ Modal logics

Topic 1.1

Propositional logic: Syntax

An example beyond good and evil

In a country, every person is either good or evil. Good people always tell the truth and evil people always lie.

You meet two citizens A and B . A says, "I am evil or B is good".

What are A and B ? **Exercise 1.1: Work this out! How do we find all possibilities?**

This lecture: How to formalize this reasoning?

- ▶ Step 1: Introduce symbols!
 - ▶ Let p_A denote the statement: A is good. It is either true or false.
 - ▶ Similarly, let p_B denote " B is good."
- ▶ Step 2: Encode puzzle using the symbols:
$$\underbrace{(\neg p_A \vee p_B)}_{\text{Statement of A}} \Leftrightarrow p_A$$
- ▶ Step 3: Solve the puzzle: Assign truth values to p_A and p_B such that the above "formula" holds. (how? trial and error?)

From English to Math

So in this example we went from an English puzzle to a symbolic logic problem.

What does a language have?

- ▶ Symbols: good, A, B, is, not
- ▶ Syntax: Grammar
- ▶ Semantics: Meaning

But any natural language has so much ambiguity!

“I saw the man with the telescope.”

A Symbolic logic

The logic is over a list of **propositions**. Some examples:

- ▶ A is good. p_A
- ▶ This tree is green q
- ▶ The fridge is broken r
- ▶ ... *many more*
 - ▶ A proposition: any sentence for which we can answer if it is True or False.
 - ▶ We do not care what each one says: we can give them a symbol.
 - ▶ Not a proposition: "May the force be with you!"

Exercise 1.2

- ▶ Give two examples of sentences that are not propositions.
- ▶ Give three examples of propositions.

Propositions as Boolean variables

We start with a set **AP** of **finite or countably many** (atomic) propositions.

- ▶ Since **AP** is countable, we assume that they are **indexed**.

$$\mathbf{AP} = \{p_1, p_2, \dots\}$$

- ▶ These are just **names/symbols** without inherent meaning
- ▶ We may also use $p, q, r, \dots, x, y, z$ to denote them
- ▶ Propositions are **Boolean variables** (as they take true or false as value!)

Topic 1.2

Propositional Formulas

Logical connectives to build propositions

Starting from “atomic” propositions, we can combine them as follows:

Building new propositions from existing ones

- ▶ a statement that says **negation** of another, e.g., *A is not good*.
- ▶ two statements are true **together**, e.g., *A is good & it is raining*.
- ▶ **at least one of the two** statements are true, e.g., *A is good or it is raining*.
- ▶ Implication: **if** A is true **then** B is also true, e.g., *If it rains then trees will grow*.
- ▶ Equivalence: truth value of two statements are **same**, e.g., *water is blue if and only if sky is black*.

We need True and False also!

- ▶ **always true**: e.g., *If A is good, then A is good*.
- ▶ **always false**: e.g., *If A is good, then A is bad*.

Logical connectives

formal name	symbol	read as	
true	$\top$	top	} 0-ary symbols
false	$\perp$	bot	
negation	$\neg$	not	} unary symbols
conjunction	$\wedge$	and	
disjunction	$\vee$	or	} binary symbols
implication	$\rightarrow$	implies	
equivalence	$\leftrightarrow$	iff	
exclusive or	$\oplus$	xor	
open parenthesis	(		} punctuation
close parenthesis	)		

- ▶ We assume that the logical connectives are not in **AP**.
- ▶ Do we need all these connectives? Which can be derived from others?

Propositional formulas

A propositional formula is a **finite string** containing symbols in **AP** and logical connectives. The set of all propositional formulas over **AP** is denoted **Prop**. Well, not every string...

Formal Definition (Note! This is an inductive definition)

The set of propositional formulas is the smallest set **Prop** such that

- ▶ $\top, \perp \in \mathbf{Prop}$
- ▶ if $p \in \mathbf{AP}$ then $p \in \mathbf{Prop}$
- ▶ if $F \in \mathbf{Prop}$ then $\neg F \in \mathbf{Prop}$
- ▶ if $\circ$ is a binary symbol and $F, G \in \mathbf{Prop}$ then $(F \circ G) \in \mathbf{Prop}$

Propositional formulas

Another way to write this definition . This is called the **Backus Naur Form**.

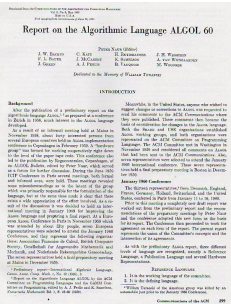
 $F \in \mathbf{Prop}$ if

$$F ::= p \mid \top \mid \perp \mid \neg F \mid (F \vee F) \mid (F \wedge F) \mid (F \rightarrow F) \mid (F \leftrightarrow F) \mid (F \oplus F)$$

where $p \in \mathbf{AP}$. Any such formula is also called **well-formed**.

A bit of history

Named after John Backus and Peter Naur, who used it to formally specify the grammar of ALGOL 60 in 1963 – one of the earliest programming languages, and ancestor of Pascal, C, and more!



Examples of propositional formulas

Exercise 1.3 Which of the following belong to **Prop** (i.e., are well-formed)?

1. $\top \rightarrow \perp$
2. $(\top \rightarrow \perp)$
3. $(p_1 \rightarrow \neg p_2)$
4. (p_1)
5. $\neg\neg\neg\neg\neg\neg\neg\neg p_1$

Obs: Not all strings over **AP** and logical connectives are in **Prop**.

How can we argue that a string does or does not belong to **Prop**?

The Lady Or the Tiger

A prisoner must choose between three rooms. One room has a Lady and other two have tigers. If the prisoner chooses the room with the Lady, he wins! If he chooses a wrong room, he (probably) gets eaten by the tiger!

Each room has a sign in front of it. BUT Only one of the signs is true; the other two are false.

(Room1) *The Lady is not in this room*

(Room2) *The Lady is not in this room*

(Room3) *The Lady is in Room 2*

Which room has the Lady?!

Exercise 1.4

1. Solve the above puzzle!
2. Write each of the above statements in propositional logic.
3. Reduce the problem to a question in this logic!

Formalizing Exercise 1.4

- ▶ Propositions $S1, S2, S3$ represent whether the sign of room 1, 2, 3 (resp.) is true.
- ▶ Propositions $L1, L2, L3$ represent Lady in room 1, 2, 3 resp.
- ▶ The statements given:
 - ▶ $S1 \leftrightarrow \neg L1, S2 \leftrightarrow \neg L2, S3 \leftrightarrow L2$.
 - ▶ $(L1 \wedge \neg L2 \wedge \neg L3) \vee (L2 \wedge \neg L1 \wedge \neg L3) \vee (L3 \wedge \neg L1 \wedge \neg L2)$.
 - ▶ $(S1 \wedge \neg S2 \wedge \neg S3) \vee (S2 \wedge \neg S1 \wedge \neg S3) \vee (S3 \wedge \neg S1 \wedge \neg S2)$.
- ▶ Is there a way to give truth values to propositions such that all these constraints are satisfied?
- ▶ **Bonus:** Can you encode this using only 3 propositions instead of 6?

Bonus: Encoding with only 3 propositions

Intuition

$S1, S2, S3$ were never really needed – they're just names for what the signs say. We can substitute their definitions directly wherever they're used.

- ▶ exactly one of $\neg L1, \neg L2, L2$ holds (exactly one sign is true)

$$(\neg L1 \wedge L2 \wedge \neg L2) \vee (L1 \wedge \neg L2 \wedge \neg L2) \vee (L1 \wedge L2 \wedge L2)$$

- ▶ exactly one of $L1, L2, L3$ holds (Lady is in exactly one room)

$$(L1 \wedge \neg L2 \wedge \neg L3) \vee (L2 \wedge \neg L1 \wedge \neg L3) \vee (L3 \wedge \neg L1 \wedge \neg L2)$$

Another Bonus: Can we encode it with just two propositions?